Critical paper review

How to read a paper and write a critique

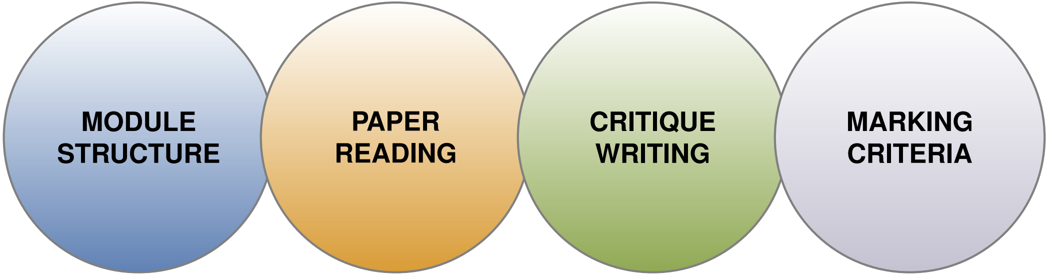
Dr. Ken Pierce
kenneth.pierce@ncl.ac.uk

School of Computing Science

Newcastle University

CSC2034 — Contemporary Topics in Computing
2026-02-09

Outline


MODULE STRUCTURE
PAPER READING
CRITIQUE WRITING
MARKING CRITERIA

PART I

Module structure

General lectures

1. Module overview
2. Assignment 1: reflective log
3. **Assignment 2: article review**
4. Finding and reading academic sources [L]
5. Assignment 3: project
6. Finding information for your project [L]
7. Technical writing [L]
8. Research chats (TBC)

   Easter break

9. Research chats (TBC)
10. Typesetting reports with LaTeX

FDC G.41

every **Monday** @14:30

Coursework

- **Reflective log** (20%) — due immediately after taster weeks
  - one entry per topic (≈ 256 words)
  - reflection on the experience and lessons learned
- **Article review** (20%) — due in the middle of March
  - structured critical review (max. 1*k* words)
  - three articles per topic to choose from
  - project independent
- **Project** (60%) — due in the first week of May
  - mini-dissertation / report format (max. 2.5*k* words)
  - specific task in one of the topics, assigned based on preferences

Coursework

- **Reflective log** (20%) — due immediately after taster weeks
  - one entry per topic ($\approx$ 256 words)
  - reflection on the experience and lessons learned
- **Article review** (20%) — due in the middle of March
  - structured critical review (max. 1k words)
  - three articles per topic to choose from
  - project independent
- **Project** (60%) — due in the first week of May
  - mini-dissertation / report format (max. 2.5k words)
  - specific task in one of the topics, assigned based on preferences

Coursework

- **Reflective log** (20%) — due immediately after taster weeks
  - one entry per topic ($\approx$ 256 words)
  - reflection on the experience and lessons learned
- **Article review** (20%) — due in the middle of March
  - structured critical review (max. 1k words)
  - three articles per topic to choose from
  - project independent
- **Project** (60%) — due in the first week of May
  - mini-dissertation / report format (max. 2.5k words)
  - specific task in one of the topics, assigned based on preferences

PART II

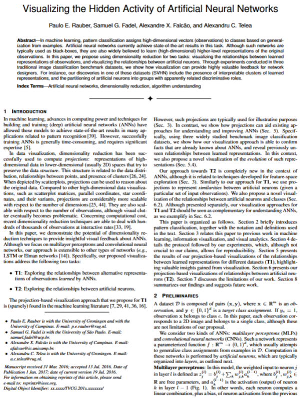
Paper reading

What is in a paper?

A scientific paper is a formal document that presents original research findings and contributes to the body of knowledge in a specific field of study.

1. Abstract
2. Introduction
3. Methods
4. Results
5. Discussion
6. Summary

Headers and order will vary depending on domain/venue



Visualizing the Hidden Activity of Artificial Neural Networks

Paulo E. Rauber, Samuel G. Fadel, Alexandre X. Falcão, and Alexandru C. Telea

Abstract—In machine learning, pattern classification assigns high-dimensional vectors (observations) to classes based on generalization from examples. Artificial neural networks currently achieve state-of-the-art results in this task. Although such networks are typically used as black-boxes, they are also widely believed to learn (high-dimensional) higher-level representations of the original observations. In this paper, we propose using dimensionality reduction for two tasks: visualizing the relationships between learned representations of observations, and visualizing the relationships between artificial neurons. Through experiments conducted in three traditional image classification benchmark datasets, we show how visualization can provide highly valuable feedback for network designers. For instance, our discoveries in one of these datasets (SVHN) include the presence of interpretable clusters of learned representations, and the partitioning of artificial neurons into groups with apparently related discriminative roles.

Index Terms—Artificial neural networks, dimensionality reduction, algorithm understanding

1 INTRODUCTION

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

Tips for academic reading

DOs

- Read effectively, efficiently, and strategically
- **Skim** and **scan** texts before reading in depth
- Exercise **critical thinking** when researching
- Focus on authentic, authoritative, and **academic information**

DON'Ts

- Try to read everything in order like a novel
- Trust everything you read without thought
- Rely on large language models — they hallucinate

Tips for academic reading

DOs

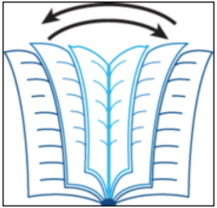
- Read effectively, efficiently, and strategically
- **Skim** and **scan** texts before reading in depth
- Exercise **critical thinking** when researching
- Focus on authentic, authoritative, and **academic information**

DON'Ts

- Try to read everything in order like a novel
- Trust everything you read without thought
- Rely on large language models — they hallucinate

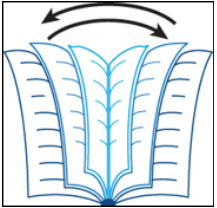
Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a ‘big picture’
- It’s **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



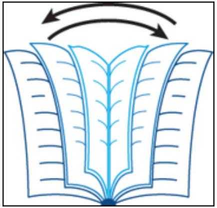
Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a ‘big picture’
- It’s **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



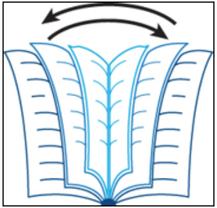
Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a ‘big picture’
- It’s **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



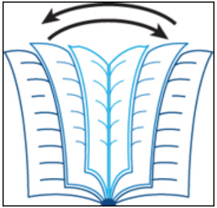
Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a ‘big picture’
- It’s **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



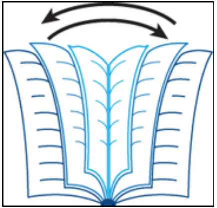
Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a 'big picture'
- It's **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



Skimming

- A **superficial, fast** activity — flicking through a book
- Acquire a ‘big picture’
- It’s **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
  - Titles, subtitles, contents pages, indexes, headings, ...
  - Figures, diagrams, plots
  - Abstracts, introductions, and conclusions
  - Topic sentences
- **Be fast. Scroll. Annotate.**



Scanning

1. Look for the **gist** of paragraphs
2. Identify key **sites of meaning** in sentences
3. Look for **topic**, **action**, and **detail**:

   Despite its popularity, social media has become an insidious, negative and highly addictive means of disseminating 'fake news.'
4. Elements
   - Topic: social media
   - Action: disseminating
   - Detail: fake news

Scanning

1. Look for the **gist** of paragraphs
2. Identify key **sites of meaning** in sentences
3. Look for **topic**, **action**, and **detail**:

   Despite its popularity, social media has become an insidious, negative and highly addictive means of disseminating 'fake news.'

4. Elements
   - Topic: social media
   - Action: disseminating
   - Detail: fake news

Scanning

1. Look for the **gist** of paragraphs
2. Identify key **sites of meaning** in sentences
3. Look for **topic**, **action**, and **detail**:

   Despite its popularity, social media has become an insidious, negative and highly addictive means of disseminating 'fake news.'

4. Elements
   - Topic: social media
   - Action: disseminating
   - Detail: fake news

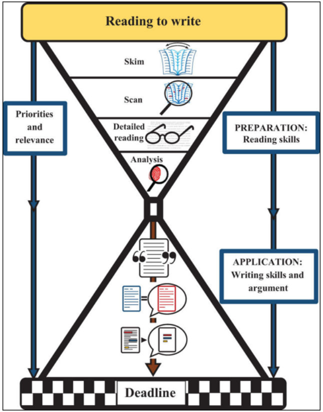
Scanning

1. Look for the **gist** of paragraphs
2. Identify key **sites of meaning** in sentences
3. Look for **topic**, **action**, and **detail**:

   Despite its popularity, social media has become an insidious, negative and highly addictive means of disseminating 'fake news.'
4. Elements
   - Topic: social media
   - Action: disseminating
   - Detail: fake news

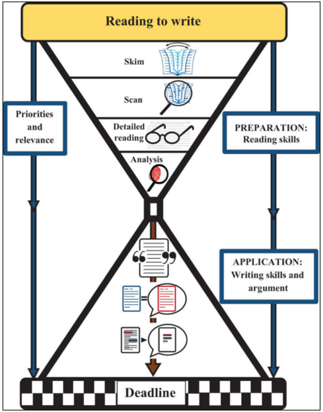
Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



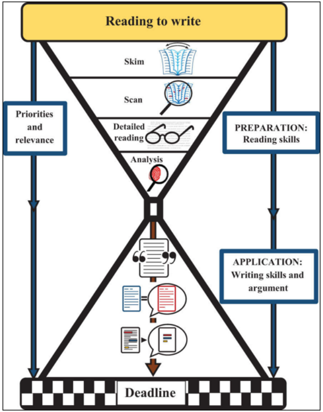
Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



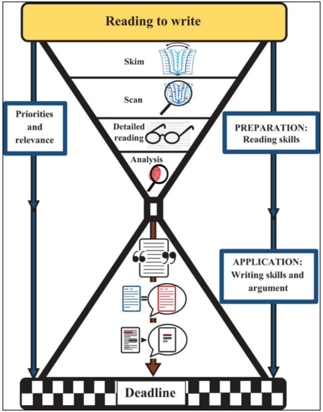
Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



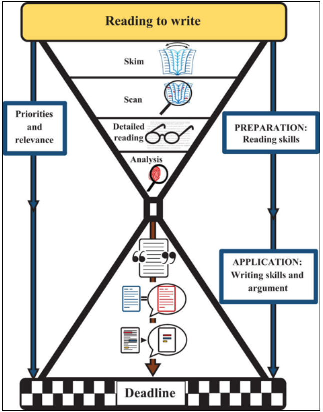
Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



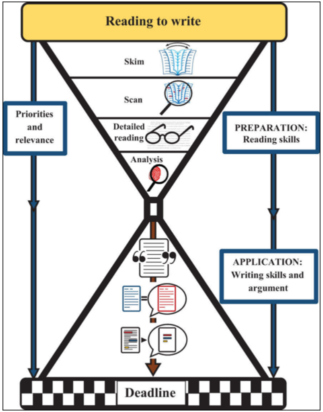
Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



Reading to write

1. Reading to understand vs **reading to write**
2. Both important skills to practice
3. Governed by other issues:
   - Time
   - Relevance
   - Priorities
4. Paraphrasing and summarising
5. Set time limits
6. Return to detail once you have 'big picture'



Part III

Critique writing

Writing the review

- Canvas quiz guides the review
- Nine questions covering the following:
  1. Which paper you picked
  2. Introduction
  3. Objectives
  4. Methods
  5. Results
  6. Discussion
  7. Research context
  8. Summary
  9. Reference

criticise
verb

- indicate the faults of something in a disapproving way.
- **form and express a judgement of a scientific or artistic work.**

Thinking about research context

The 'W' questions

- **Who** conducted the research? What were/are their interests?
- **When** and **where** was the research conducted?
- **Why** did they do this research?
- Does the research have a broad (even global) relevance?

The context

- Were others pursuing **related research** at the time the work was done?
- Who funded the research? Did the **funder** influence the research?
- On what **prior observations** was the research based?
- How **important** was the research question posed by the researcher?

Thinking about research context

The ‘W’ questions

- **Who** conducted the research? What were/are their interests?
- **When** and **where** was the research conducted?
- **Why** did they do this research?
- Does the research have a broad (even global) relevance?

The context

- Were others pursuing **related research** at the time the work was done?
- Who funded the research? Did the **funder** influence the research?
- On what **prior observations** was the research based?
- How **important** was the research question posed by the researcher?

Thinking about research significance

The effect of the research

- How did other researchers view the **significance** of the research?
- Did the research lead to **new questions** or hypotheses?
- Have other researchers **supported** or refuted the research?
- Did the research make a significant **contribution** to human knowledge?
- Did the research produce any **practical** applications?
- What are the social, political, technological implications?

Look at articles that cite this work

Part IV

Marking criteria

Requirements

You should:

- select **one** of the articles and **read** it carefully
- follow the guidance to help with **analysis**
- answer the questions to complete your review

Marking criteria:

- Content and style — relevant, accurate, error free
- Summarising — present idea, comment on organisation and presentation
- Analysis — complexity, perceptiveness, originality, and depth of though
- Use of evidence — focus, coherence, insightful,

Summary

Critical review assessment:

- gives you experience **reading** about **technical subjects**
- lets you **engage** with one of the topics further
- help you to practice **critically evaluating** and source
- gives you experience **writing** a preview of a **scientific paper**

Next

New topic intro lecture: **Human Computer Interaction**

Tommorrow

1h lecture (FDC @ **15:30**) + 2h practical session on **design requirements**

Summary

Critical review assessment:

- gives you experience **reading** about **technical subjects**
- lets you **engage** with one of the topics further
- help you to practice **critically evaluating** and source
- gives you experience **writing** a preview of a **scientific paper**

Next

New topic intro lecture: **Human Computer Interaction**

Tommorrow

1h lecture (FDC @ **15:30**) + 2h practical session on **design requirements**

References

British Dyslexia Association (2026).
How is dyslexia diagnosed?
https://www.bdadyslexia.org.uk/dyslexia/how-is-dyslexia-diagnosed/dyslexia-checklists [Accessed: 2026-02-09].

Rauber, P. E., Fadel, S. G., Falcao, A. X., and Telea, A. C. (2016).
Visualizing the hidden activity of artificial neural networks.
IEEE transactions on visualization and computer graphics, 23(1):101–110.

Wallbank, A. J. (2022).
Academic writing and dyslexia: A visual guide to writing at university.
Routledge.